

12 Hydrogen peroxide slowly decomposes at room temperature.

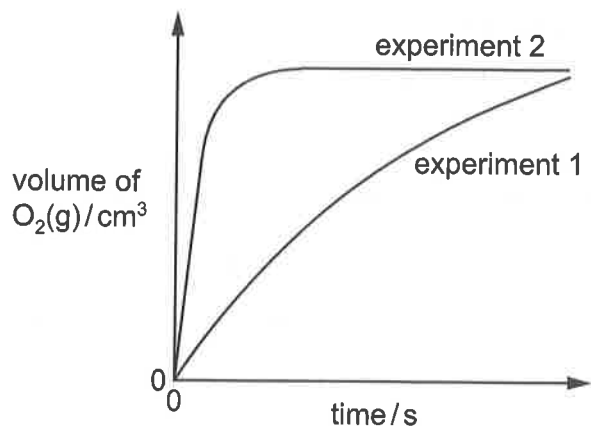


Two experiments were performed to study the effects of adding lead(IV) oxide on the decomposition of 1.5 mol dm^{-3} hydrogen peroxide at constant temperature.

Experiment 1: 20 cm^3 of hydrogen peroxide.

Experiment 2: 20 cm^3 of hydrogen peroxide and 1.0 g of lead(IV) oxide.

At the end of experiment 2, the mixture was filtered and 1.0 g of lead(IV) oxide was recovered.



Which row is correct?

	value of rate constant	activation energy
A	equal in experiment 1 and 2	higher in experiment 1
B	equal in experiment 1 and 2	higher in experiment 2
C	higher in experiment 2	higher in experiment 1
D	higher in experiment 2	higher in experiment 2